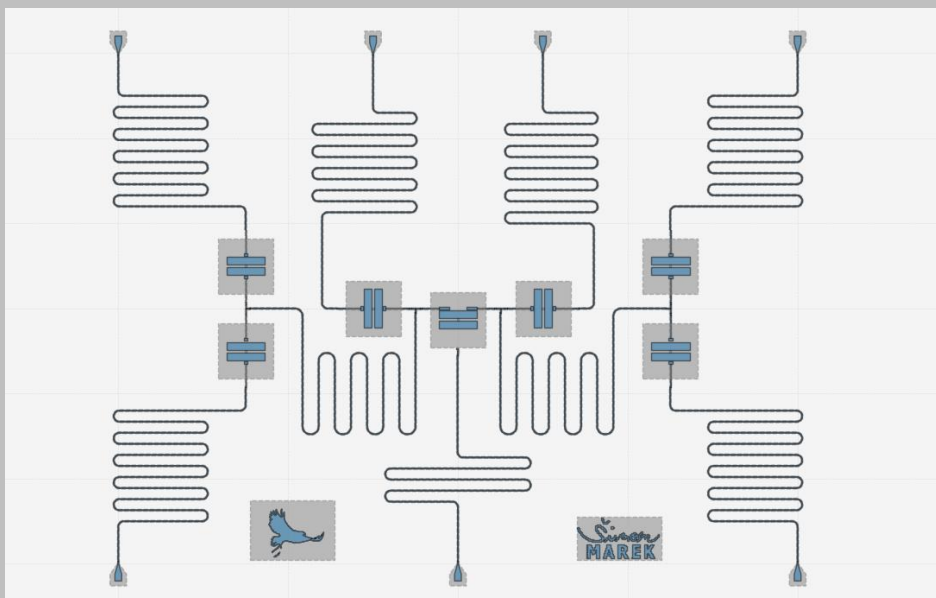
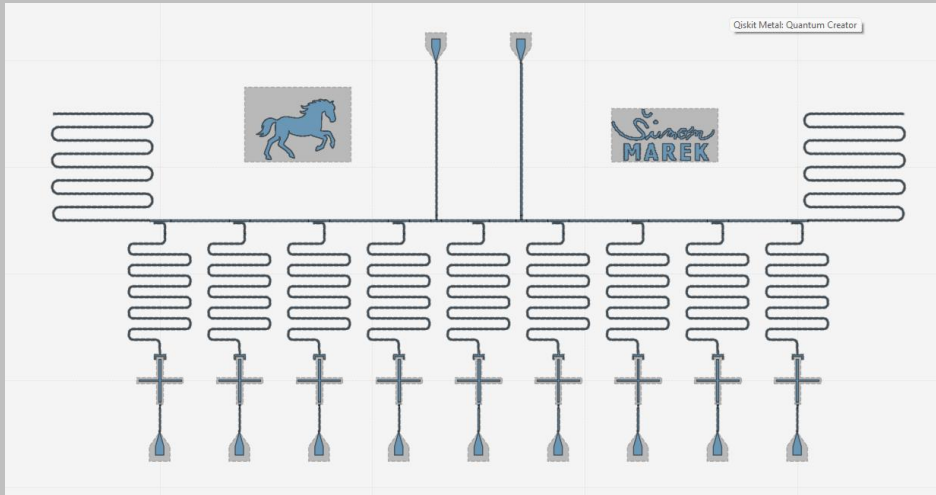
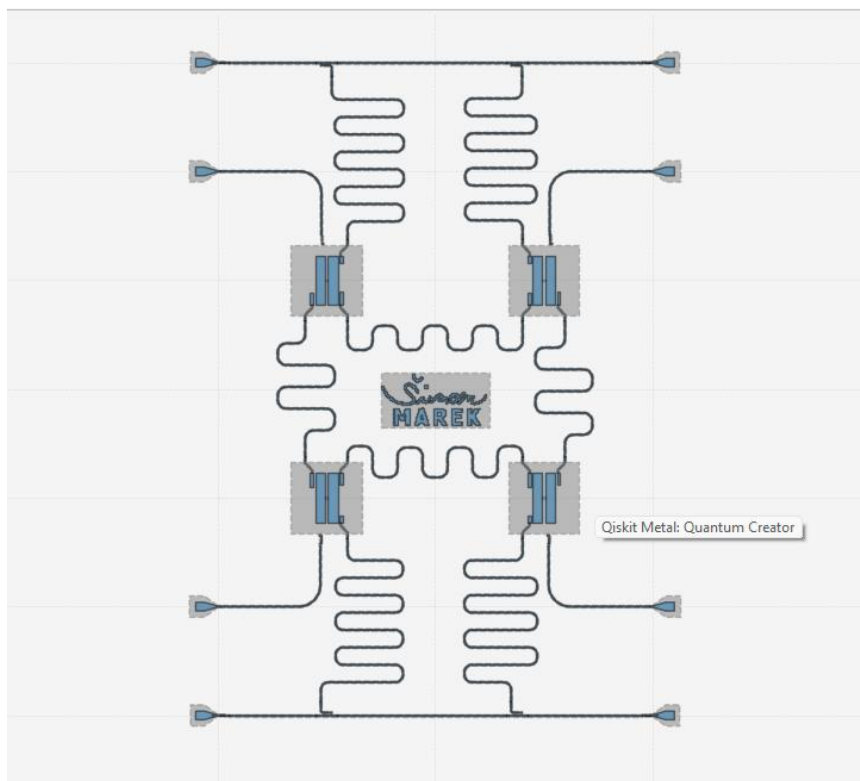


Quantum Hardware Chips

Library Reference

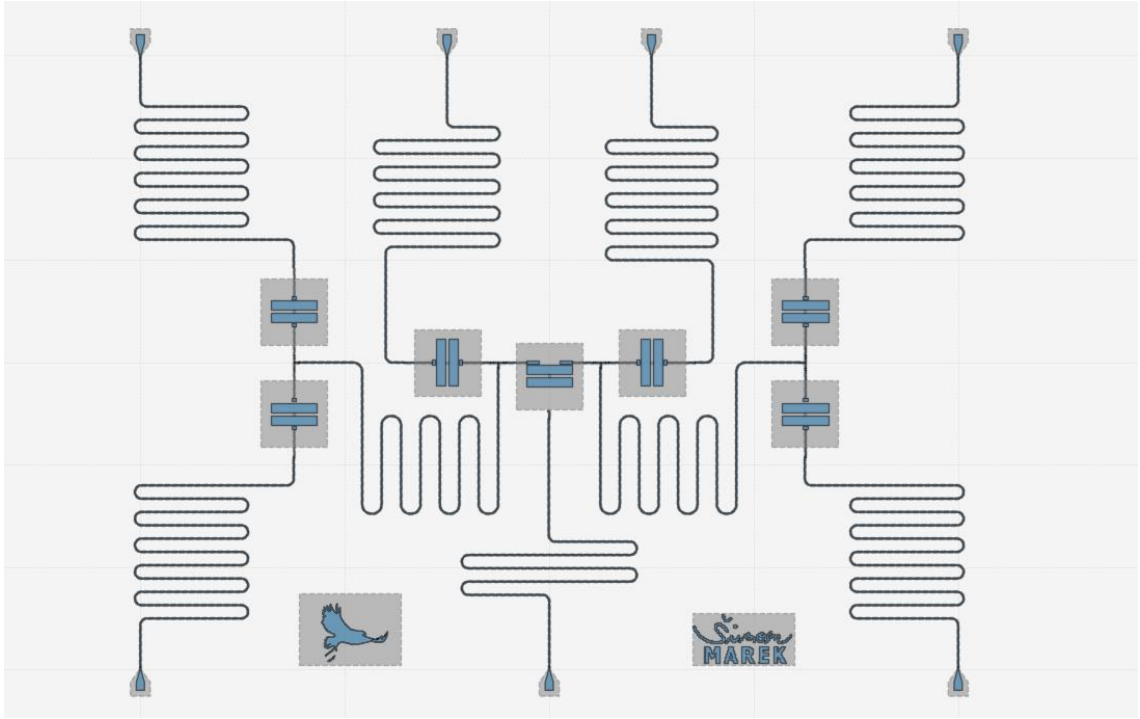




4 Qubit - Transmon Version

Simple 4 Qubit version with Transmon pockets, It has multiplied readout lines and simple coupling resonators. It can be used for evaluating simple quantum algorithms. Design is made by me, so no reference. It can be manufactured on silicon.

Built with Qiskit Metal!



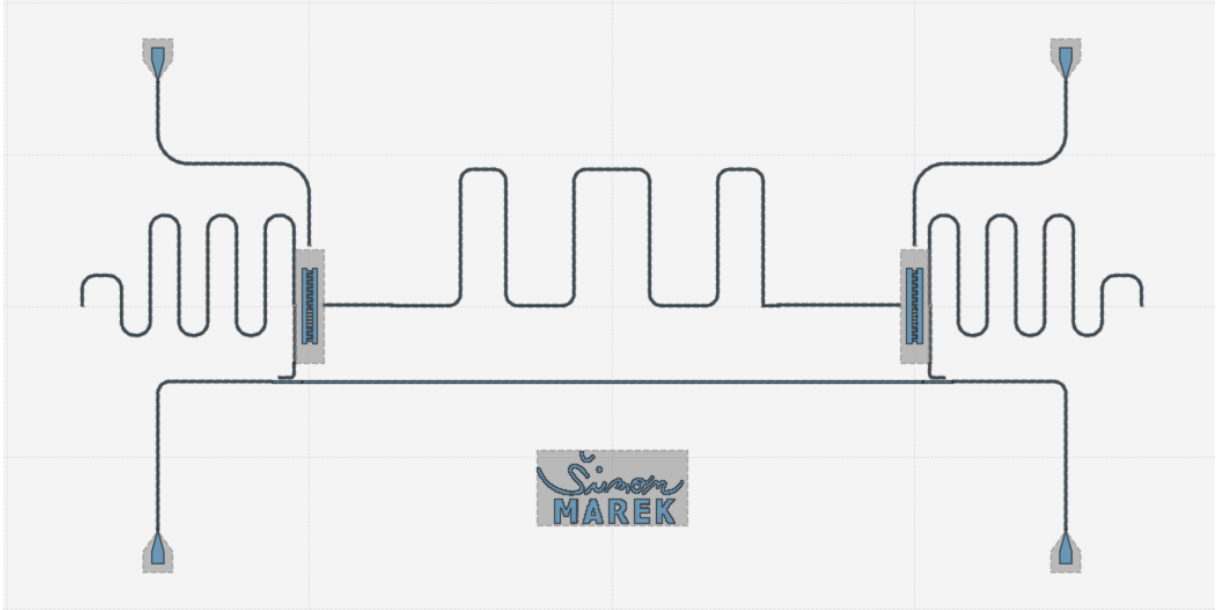
7 Qubit – Transmon Version

Middle level Quantum chip with 7 Qubits by IBM.

It uses Transmon pocket qubit! One middle flux tunable transmon with 3 ports, others with 2 ports. Single non multiplexed readout lines.

Flux bias to middle Transmon.

Built with Qiskit Metal!



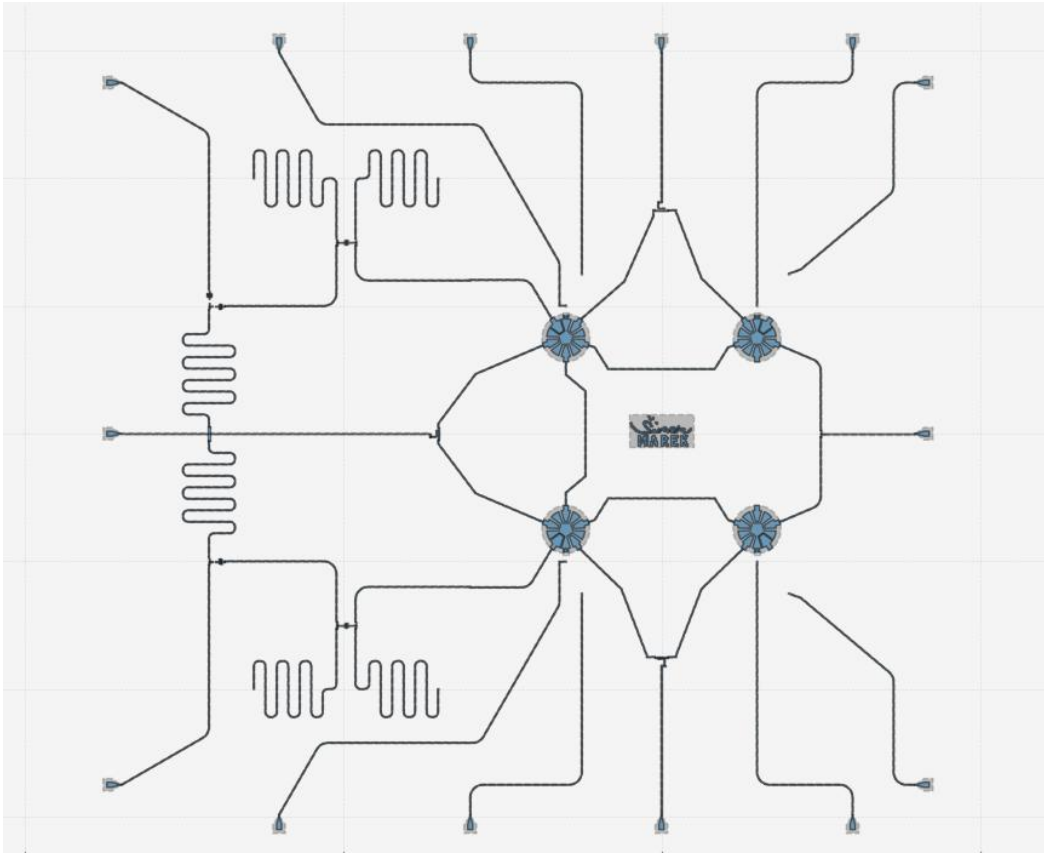
2 Qubit – Interdigitated Transmon

2 Qubit Interdigitated Transmon Qubit. With multiplexed readout resonator with dead ends used to tune at certain frequency.

Two qubits coupled together using middle coupling bus. This chip can be used for simple demonstration algorithms.

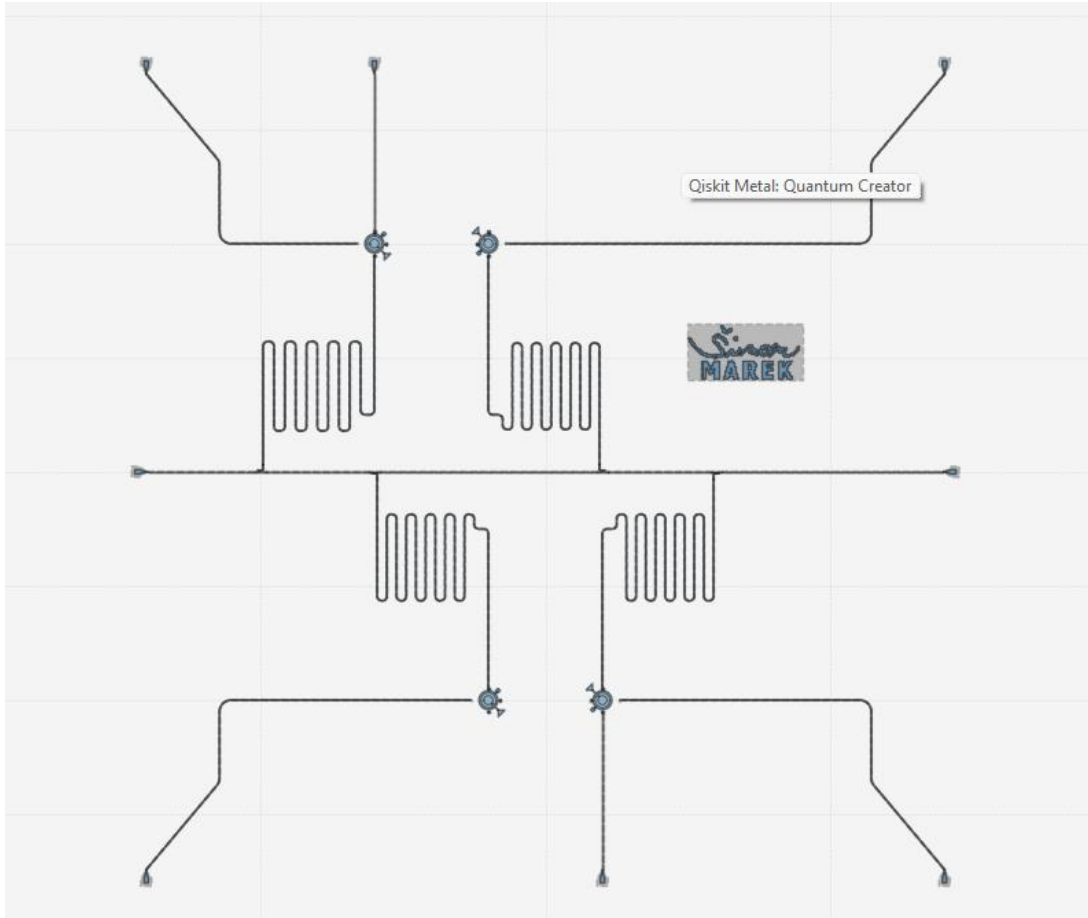
Developed at Delft.

Built with Qiskit Metal



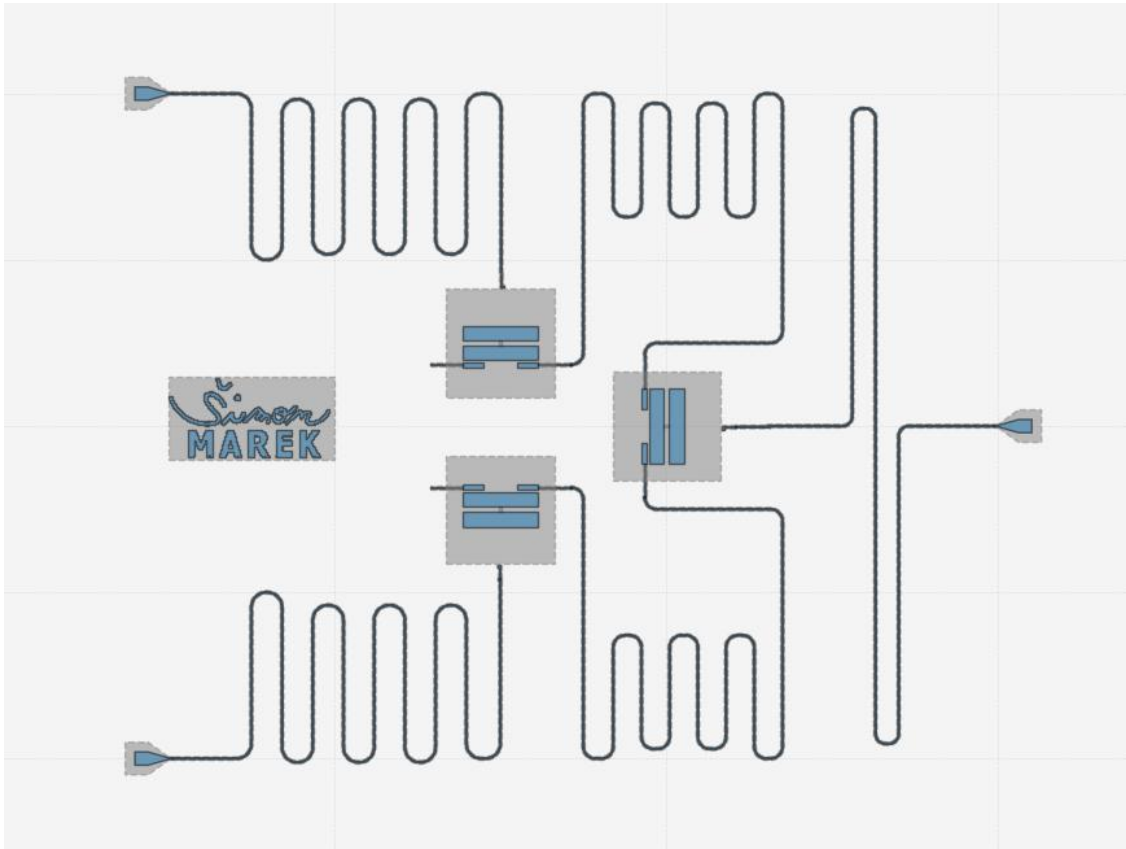
4 Qubit – Starmon Version

4 Qubit Starmon Quantum chip developed at ETHZ. It uses capacitors for purcell filtering. It has multiplexed readout lines, and additional flux biasing. Starmon is improved compared to transmon; it has more channels and is easier to use. Built with Qiskit Metal.



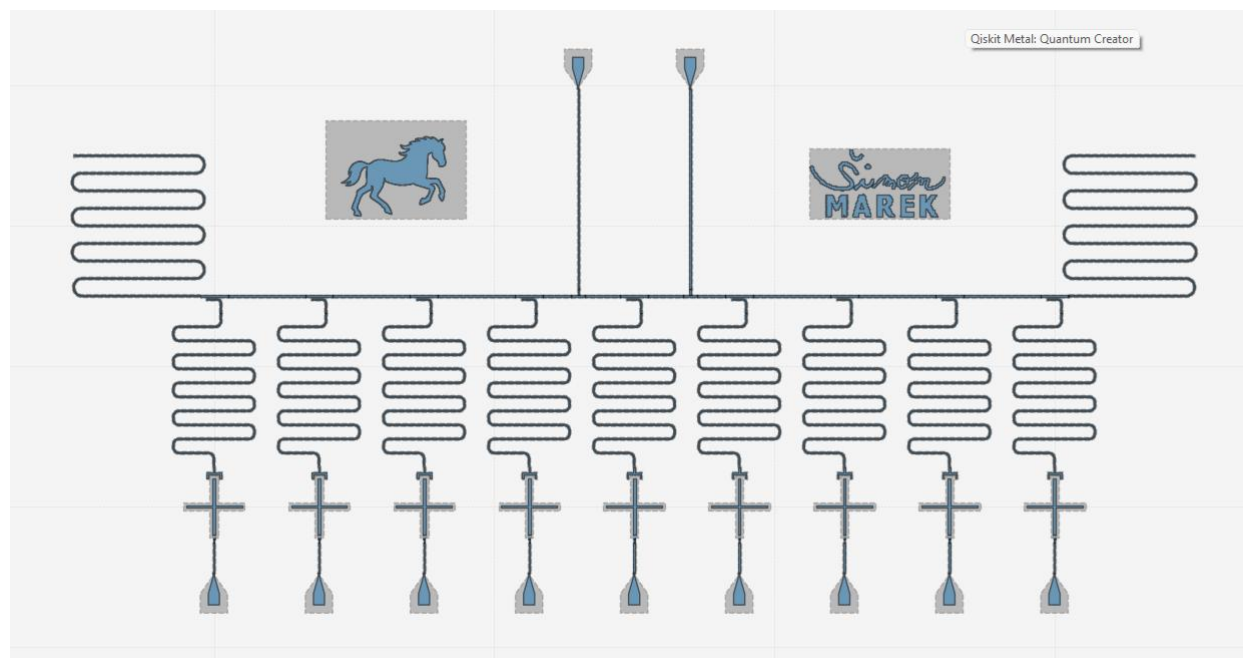
4 Qubit – Coecentric Transmon Version

4 Qubit Coecentric – Non standard qubit type. You can See flux bias on two qubits, microwave drive line and multiplexed readout lines on all 4 qubits. Since it is simple architecture it can be used only on limited number of algorithms
Build with Qiskit Metal



3 Qubit – Transmon Version

3 Qubit Transmon Pocket demonstration inspired by IBM. You can see three port devices, two qubits have occupied only 2 port and middle qubit has flux tuning. You can see non multiplexed readouts and casual coupling buses between buses
Built with Qiskit Metal



9 Qubit – Cross Transmon Version

9 Qubit Cross Transmon pocket inspired by Google and UCSB. It uses resonators put to ground to fine tune it at certain frequency. It has multiplexed readout line for all 9 qubits. It uses flux biasing for each josephson junction of Cross transmon. Can be used for more complex tasks since its bigger chip with scalability.

Built with Qiskit Metal

About author



Simon is young quantum chip leader, he pioneered visual presentation of quantum chips using renders and help grow open source quantum chip design community. He works as Quantum chip designer at Cadence. His work was recognised by world quantum hardware experts from all over the world.